Math 292 Final Formula Sheet — Page 1

First-Order ODEs (Ch 1)

1. First-Order Linear
Standard Form:

$$x'(t) + p(t)x(t) = q(t)$$

Step 1: Get Integrating Factor

$$\mu(t) = e^{\int p(t)dt}$$

Step 2: Multiply through by $\mu(t)$

$$\frac{d}{dt}(x(t)\mu(t)) = \mu(t)q(t)$$

Step 3: Integrate (Don't forget $+C$)

$$x(t) = \frac{1}{\mu(t)} \int \mu(t)q(t)\ dt$$

2. Bernoulli Equations
Standard Form:

$$x'(t) + p(t)x(t) = q(t)x^n(t)$$

Step 1: Divide by x^n and substitute

$$y = x^{1-n}$$
$$y' = (1-n)x^{-n} \cdot x'$$

Step 2: Solve the new linear ODE

$$y' + (1-n)p(t)y = (1-n)q(t)$$

3. Riccati Equations
Standard Form:

$$x'(t) = p(t) + q(t)x(t) + r(t)x^2(t)$$

Step 1: Guess particular solution

$$x_p(t) = \alpha t^r$$

Step 2: Substitute and solve for $u(t)$

$$x(t) = x_p(t) + u(t)$$

Lipschitz Condition (Ch 2)

Given

$$v(x) = x'(t) : \forall x \in \mathbb{R}$$

Step 1: If $v'(x)$ is bounded, $v(x)$ is Lipschitz:

$$|v(x) - v(y)| \leq L|x - y|$$

Step 2: Check

$$\lim_{x\to\infty} v'(x) < \infty$$

If so, $v(x)$ is closed and bounded for large $N \in \mathbb{N}$.
Step 3: By EVT, we have a max

$$|v'(x)| \leq k : \forall x \in [-N, N].$$

Step 4: By MVT, we have our Lipschitz constant

$$L = |v'(x)| : \forall x \in \mathbb{R}$$

Matrix Exponential (Ch 3)

1. Diagonalizable

$$e^{tA} = Ve^{tD}V^{-1}$$

2. Defective

$$e^{tA} = e^{t\lambda}(I + t(A - \lambda I))$$

Extra
For $\mathbf{x}'(t) = A\mathbf{x}(t)$:

$$\mathbf{x}(t) = C_1\mathbf{v_1}e^{\lambda_1 t} + C_2\mathbf{v_2}e^{\lambda_2 t}$$

Sub $t = 0$ to solve for the constants.

Duhamel's Formula (Ch 4)

$$\mathbf{x}(t) = e^{(t-t_0)A}\mathbf{x}_0 + \int_{t_0}^{t} e^{(t-s)A}\mathbf{g}(s)\ ds.$$

Picard Iteration (Ch 5)

Set $x_0(t) = x_0$. Then, iterate the following

$$x_{n+1}(t) = x_0 + \int_{t_0}^{t} f(s, x_n(s))\, ds$$

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Second-Order ODEs (Ch 6)

Standard Form:

$$x''(t) + p(t)x'(t) + q(t) = f(t)$$

Constant Coefficients (p, q)
Step 1: Solve for roots r_1, r_2

$$r^2 + pr + q = 0$$

Step 2: Solve homogeneous $x_h(t)$

$$r_1 \neq r_2 \implies x_h(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$
$$r_1 = r_2 \implies x_h(t) = c_1 e^{rt} + c_2 t e^{rt}$$

Step 3: Guess particular $x_p(t)$

$$f(t) = e^{\alpha t} \implies x_p(t) = Ae^{\alpha t}$$
$$f(t) = t \implies x_p(t) = At$$

If $x_p(t)$ is already a solution multiply by t.

Cauchy-Euler Equations
Standard Form

$$t^2 x''(t) + atx'(t) + bx(t) = f(t)$$

Step 1: Substitute $x(t) = t^r$ and solve for r.
Step 2: Solve homogeneous $x_h(t)$

$$r_1 \neq r_2 \implies x_h(t) = c_1 t^{r_1} + c_2 t^{r_2}$$
$$r_1 = r_2 \implies x_h(t) = c_1 t^r + c_1 t^r \ln|t|$$

Variation of Parameters
Step 1: Let

$$M(s) = \begin{pmatrix} x_1(s) & x_2(s) \\ x_1'(s) & x_2'(s) \end{pmatrix}$$
$$N(t,s) = \begin{pmatrix} x_1(s) & x_2(s) \\ x_1(t) & x_2(t) \end{pmatrix}$$

Step 2: Solve particular solution

$$x_p(t) = \int_{t_0}^{t} \frac{det(N(t,s))}{det(M(s))} \cdot f(s)\, ds$$

Second-Order ODEs (Ch 6)

Variation of Parameters (continued...)
Step 3: Our constants are:

$$c_1 = \frac{x(t_0)x_2'(t_0) - x'(t_0)x_2(t_0)}{det(M(t_0))}$$
$$c_2 = \frac{x'(t_0)x_1(t_0) - x(t_0)x_1'(t_0)}{det(M(t_0))}$$

Step 4: Combine for general solution

$$x(t) = c_1 x_1(t) + c_2 x_2(t) + x_p(t)$$

Metric Spaces & L^2

Supremum Distance

$$d_\infty(f,g) = \max_{x \in [a,b]} |f(x) - g(x)|$$

Step 1: Find critical points of $h = |f - g|$
Step 2: Compare $|h|$ at crit points and endpoints.

Best L^2 Approximation
Given orthogonal set $\{u_0, u_1, \ldots, u_n\}$, where

$$\hat{f} = \sum_{k=0}^{n} c_k u_k$$

Solve for the coefficients:

$$c_k = \frac{\langle f, u_k \rangle}{\langle u_k, u_k \rangle}$$

Extra Stuff

$$\frac{d}{dx}\cos(x) = -\sin(x), \quad \frac{d}{dx}\sin(x) = \cos(x)$$
$$\int te^{at}\, dt = \frac{e^{at}}{a^2}(at - 1)$$
$$\int x^2 \cos x\, dx = (x^2 - 2)\sin x + 2x\cos x$$
$$\cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin^2 x = \frac{1 - \cos 2x}{2}$$
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$